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# 1. Define the Problem

Fig12.1(a) (Figure 1.1) shows the first Shaheed Minar, a masonry structure built overnight by medical students (and soon demolished by the then-government). In Fig. 12.1(b) (Figure 1.1) it is represented by the monument ABCD composed of square-shaped cross-sections [Fig. 12.1(c)] (Figure 1.1). Assume horizontal deflections for the structure fixed at A ($u_A = 0$).

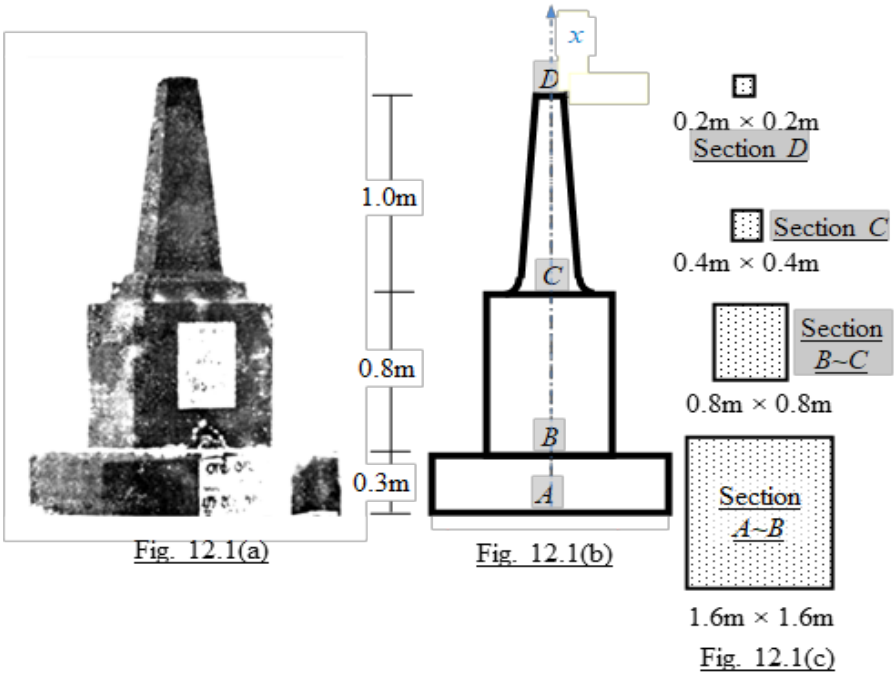



*Figure 1.1 Problem Figure*

## 1.1.Project 1

Calculate: Horizontal Stiffness ($k_{AB} + k_{BC} + k_{CD}$) using appropriate shape function $\psi(x)$ [Use Gauss Integration (n = 2)]

## 1.2.Project 2

Calculate and compare natural frequencies (for horizonal motion) of Fig.12.1 (Figure 1.1).

## 1.3.Comparison

Compare the results with,

- Many Triangular Elements,
- Many Q4 Elements

# 2. Horizontal Stiffness and Natural frequency using Shape function

Section: **A1**

Group Member Registration Number: **20105016, 22109501, 20105006**

So, $E = (10 + 0.1R_0)$ Gpa = 10+0.1*(16+01)/2 = **10.85 Gpa;**

*Here, 06 is not present because calculation of horizontal stiffness and natural frequency using shape functions was already completed before joining the team.*

## 2.1.Shape Function Assumption

There are several shape functions (Table 2-1) that could be used. Among such shape functions "Power Function $\boldsymbol{\psi(x) = \left(\frac{x}{L}\right)^2}$"(Figure 2.1) (1)

is used.

As,

Shape Function, $\psi(x) = \left(\frac{x}{L}\right)^2$

Deflection at A = 0, D=u

So,

$$\psi(x) = \left(\frac{x}{L}\right)^2;\ \psi(0) = 0; \psi(L) = 1$$

$$\psi'(x) = \left(\frac{2x}{L^2}\right);\ \psi'(0) = 0;$$

$$\psi''(x) = \frac{2}{L^2};\ \psi(0) \neq 0$$

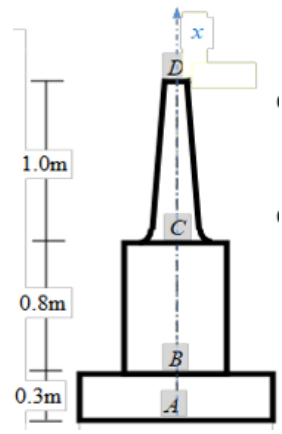



*Table 2-1: Different Types of Shape Function*

| Shape Functions | Descriptions |
|---|---|
| Sigmoid Function; $\psi(x) = \frac{1}{1+e^{-k(x-L/2)}}$; | here k is a positive constant, standard k=1 |
| Power Function; $\psi(x) = \left(\frac{x}{L}\right)^a$ | Here a =2 (standard value) increasing a makes the curve steeper |
| $1 - \cos\left(\frac{\pi x}{2L}\right)$ | |
| $\sin\left(\frac{\pi x}{2L}\right)$ | |
| Gaussian Function; $\psi(x) = e^{-\frac{\left(x-\frac{L}{2}\right)^2}{2\sigma^2}}$ | $\sigma = \frac{L}{6}(standard)$; Decreasing $\sigma$ makes the peak narrower and steeper |

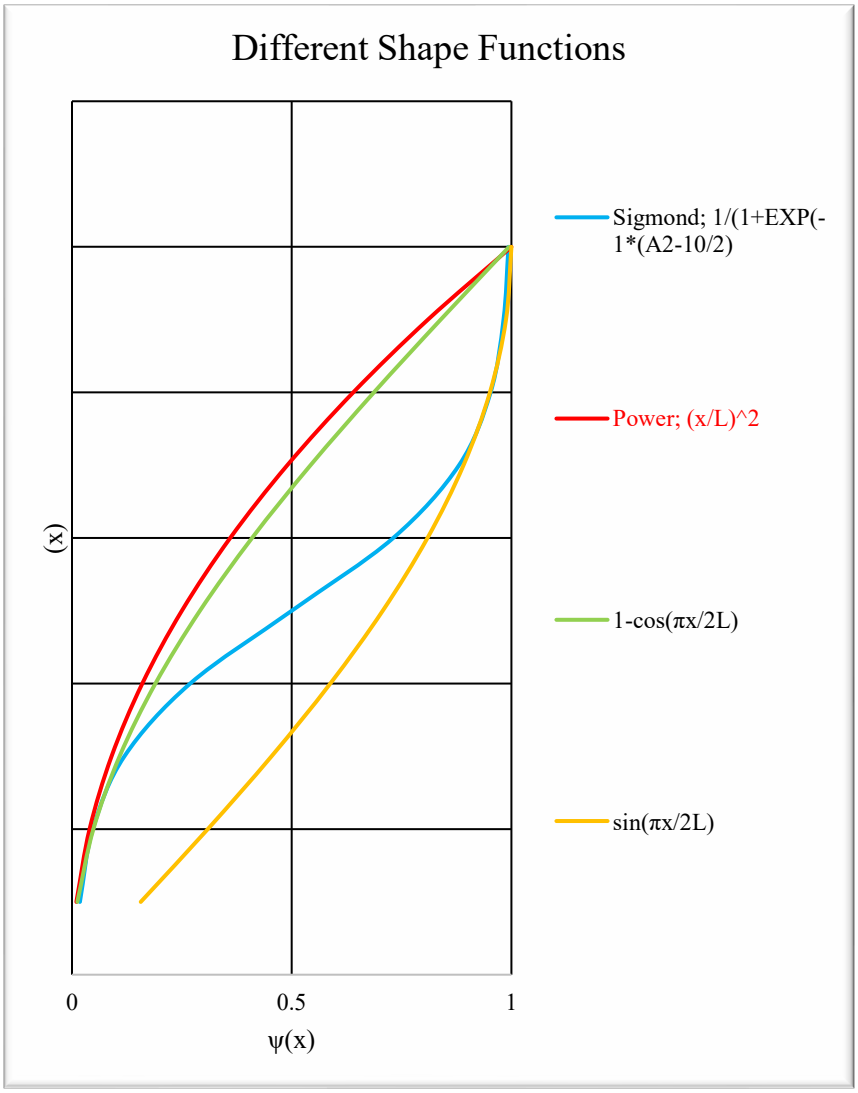



*Figure 2.1: Different Types of Shape Function*

## 2.2. Horizontal Stiffness

We know, $k^* = \int_0^L EI(\psi''(x))^2\, dx$

$$= \int_0^{0.3} EI_{AB}(\psi''(x))^2\, dx + \int_{0.3}^{1.1} EI_{BC}(\psi''(x))^2\, dx + \int_{1.1}^{2.1} EI_{CD}(\psi''(x))^2\, dx$$

Here,

$$EI_{AB} = 10.85\ GPa\ * \frac{1.6 * 1.6^3}{12} m^4 = 5.92\ * 10^9\ Nm^2$$

$$EI_{BC} = 10.85\ GPa\ * \frac{0.8 * 0.8^3}{12} m^4 = 0.37\ * 10^9\ Nm^2$$

$$EI_{CD} = E * \frac{bh^3}{12};$$

*here b and h are variable* (Figure 2.2) *so a function is needed*

$b@1.1 = 0.4$ *and* $b@2.1 = 0.2$

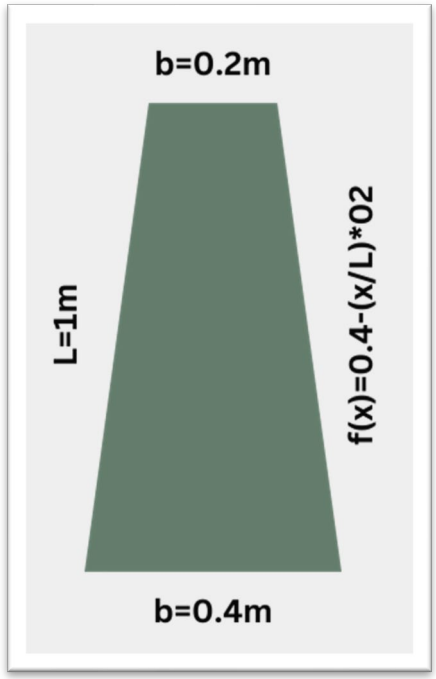



*Figure 2.2: Variable Cross-section from C to D*

So, the function of "b" should be $f(x) = 0.4 - 0.2 * \frac{x}{L}$
Same for "h"
Now,

$$k^* = \int_0^L EI(\psi''(x))^2\, dx$$

$$= \int_0^{0.3} \mathrm{EI_{AB}}(\psi''(\mathrm{x}))^2\, \mathrm{dx} + \int_{0.3}^{1.1} \mathrm{EI_{BC}}(\psi''(\mathrm{x}))^2\, \mathrm{dx} + \int_{1.1}^{2.1} \mathrm{EI_{CD}}(\psi''(\mathrm{x}))^2\, \mathrm{dx}$$

$$= 5.92 * 10^9 \int_0^{0.3} \left(\frac{2}{\mathrm{L}^2}\right)^2 \mathrm{dx} + 0.37 * 10^9 \int_{0.3}^{1.1} \left(\frac{2}{\mathrm{L}^2}\right)^2 \mathrm{dx} + 10.85 * 10^9 \int_{1.1}^{2.1} \frac{\left(0.4 - 0.2 * \frac{\mathrm{x}}{\mathrm{L}}\right)\left(0.4 - 0.2 * \frac{\mathrm{x}}{\mathrm{L}}\right)^3}{12} * \left(\frac{2}{\mathrm{L}^2}\right)^2 \mathrm{dx}$$

$$= 5.92 * 10^9 \int_0^{0.3} \left(\frac{2}{0.3^2}\right)^2 \mathrm{dx} + 0.37 * 10^9 \int_{0.3}^{1.1} \left(\frac{2}{0.8^2}\right)^2 \mathrm{dx} + 10.85 * 10^9 \int_{1.1}^{2.1} \frac{\left(0.4 - 0.2 * \frac{\mathrm{x}}{1}\right)\left(0.4 - 0.2 * \frac{\mathrm{x}}{1}\right)^3}{12} * \left(\frac{2}{1^2}\right)^2 \mathrm{dx}$$

**For Gauss Integration**

Here, n=2

So, $\xi_1 = 0.57735; \xi_2 = -0.57735; A_1 = 1; A_2 = 1$ (Figure 2.3)
We know, $\int_0^1 f(x)dx = \frac{b-a}{2}\sum A_i f(x)$

*And*

$$x_i = \frac{b+a}{2} + \frac{b-a}{2}\xi$$

| n | $\pm\xi_i$ | $A_i$ |
|---|---|---|
| 1 | 0.0 | 2.0 |
| 2 | ± 0.57735 | 1.0 |
| 3 | ± 0.77460<br>0.0 | 0.55556<br>0.88889 |
| 4 | ± 0.86114<br>± 0.33998 | 0.34785<br>0.65215 |
| 5 | ± 0.90618<br>± 0.53847<br>0.0 | 0.23693<br>0.47863<br>0.56889 |

*Figure 2.3: Required values for Gauss Integration*

**For AB[0~0.3] portion,**

$$x_1 = \frac{0.3+0}{2} + \frac{0.3-0}{2} * 0.57735 = 0.2366$$
$$x_2 = \frac{0.3+0}{2} - \frac{0.3-0}{2} * 0.57735 = 0.0634$$
$$\int_0^{0.3} f(x)\,dx = \int_0^{0.3} 1\,dx = \frac{0.3-0}{2} * (1+1) = 0.3$$

**For BC[0.3~1.1] portion,**

$$x_1 = \frac{1.1+0.3}{2} + \frac{1.1-0.3}{2} * 0.57735 = 0.9309$$
$$x_2 = \frac{1.1+0.3}{2} - \frac{1.1-0.3}{2} * 0.57735 = 0.4691$$
$$\int_{0.3}^{1.1} 1\,dx = \frac{1.1-0.3}{2} * (1+1) = 0.8$$

**For CD[1.1~2.1] portion,**

$$\int_{1.1}^{2.1} \frac{(0.4-0.2*x)(0.4-0.2*x)^3}{12} * 2^2 \, dx$$

$$= \frac{1}{3}\int_{1.1}^{2.1} (0.4-0.2*x)(0.4-0.2*x)^3 dx$$

$$x_1 = \frac{2.1+1.1}{2} + \frac{2.1-1.1}{2} * 0.57735 = 1.8887$$

$$x_2 = \frac{2.1+1.1}{2} - \frac{2.1-1.1}{2} * 0.57735 = 1.3113$$

$$\frac{1}{3}\int_{1.1}^{2.1} (0.4-0.2*x)(0.4-0.2*x)^3 dx$$

$$= \frac{1}{3} * \frac{2.1-1.1}{2} * \begin{bmatrix} 1*\{(0.4-0.2*1.8887)(0.4-0.2*1.8887)^3\} \\ +1*\{(0.4-0.2*1.3113)(0.4-0.2*1.3113)^3\} \end{bmatrix}$$

$$= 6*10^{-5}$$

Now, $k^* = \int_0^L EI(\psi''(x))^2 \, dx$

$$= \int_0^{0.3} EI_{AB}(\psi''(x))^2 \, dx + \int_{0.3}^{1.1} EI_{BC}(\psi''(x))^2 \, dx + \int_{1.1}^{2.1} EI_{CD}(\psi''(x))^2 \, dx$$

$$= 5.92 * 10^9 \int_0^{0.3} \left(\frac{2}{0.3^2}\right)^2 dx + 0.37 * 10^9 \int_{0.3}^{1.1} \left(\frac{2}{0.8^2}\right)^2 dx + 10.85$$

$$* 10^9 \int_{1.1}^{2.1} \frac{\left(0.4-0.2*\frac{x}{1}\right)\left(0.4-0.2*\frac{x}{1}\right)^3}{12} * \left(\frac{2}{1^2}\right) dx$$

$$= 5.92 * 10^9 * \left(\frac{2}{0.3^2}\right)^2 * 0.3 + 0.37 * 10^9 * \left(\frac{2}{0.8^2}\right)^2 * 0.8 + 10.85 * 10^9 * 6 * 10^{-5}$$

$$= 8.799 * 10^{11} \frac{N}{m} = 8.799 * 10^8 \frac{kN}{m}$$

So, horizontal Stiffness using shape function is,

$$\mathbf{k^* = 8.799 * 10^9 \frac{kN}{m}}$$

## 2.3. Natural Frequency

We know, $Natural\ Frequency,\ \omega = \sqrt{\frac{k^*}{m^*}}$

$$m^* = \int_0^L \mu(\psi(x))^2 \, dx$$

$$= \left(\int_0^L \mu\big(\psi(x)\big)^2 dx\right)$$

$$= \left( \int_0^{0.3} \mu_{AB}\left(\psi(x)\right)^2 dx + \int_{0.3}^{1.1} \mu_{BC}\left(\psi(x)\right)^2 dx + \int_{1.1}^{2.1} \mu_{CD}\left(\psi(x)\right)^2 dx \right)$$

$\mu = \gamma * \frac{\text{area}}{\text{g}}$;

*here*, $\gamma =$ unit weig*h*t of material $\left(\text{masonry } h\text{ere } 22\ \frac{kN}{m^3}\right)$

Area is X − sectinal Area, $\text{g} = 9.8\ \frac{\text{m}}{\text{s}^2}$

**For AB[0~0.3] portion,**

$$\int_0^{0.3} \mu_{AB}\left(\psi(x)\right)^2 dx$$

$$= \int_0^{0.3} \gamma * \frac{A_{AB}}{9.8} * \left(\left(\frac{x}{0.3}\right)^2\right)^2 dx$$

$$= \int_0^{0.3} 22 * \frac{1.6^2}{9.8} * \left(\left(\frac{x}{0.3}\right)^2\right)^2 dx$$

$$= 22 * \frac{1.6^2}{9.8} * \frac{1}{0.3^4} * \int_0^{0.3} x^4 dx$$

$$= 709.5 * \frac{0.3-0}{2} * ((0.2366)^4 + (0.0634)^4) = 0.335\text{kN}\frac{\text{s}^2}{\text{m}}$$

**For BC[0.3~1.1] portion,**

$$\int_{0.3}^{1.1} \mu_{BC}\left(\psi(x)\right)^2 dx$$

$$= \int_{0.3}^{1.1} \gamma * \frac{A_{BC}}{9.8} * \left(\left(\frac{x}{0.8}\right)^2\right)^2 dx$$

$$= \int_{0.3}^{1.1} 22 * \frac{0.8^2}{9.8} * \left(\left(\frac{x}{0.8}\right)^2\right)^2 dx$$

$$=22 * \frac{0.8^2}{9.8} * \frac{1}{0.8^4} * \int_{0.3}^{1.1} x^4\ dx$$

$$= 3.5 * \frac{1.1 - 0.3}{2} * (0.9309^4 + 0.4691^4)$$

$$= 1.12\ \text{kN}\frac{\text{s}^2}{\text{m}}$$

**For CD[1.1~2.1] portion,**

$$\int_{1.1}^{2.1} \mu_{CD}\left(\psi(x)\right)^2 dx$$

$$= \int_{1.1}^{2.1} \gamma * \frac{A_{CD}}{9.8}\left(\left(\frac{x}{1}\right)^2\right)^2 dx$$

$A_{CD} \Rightarrow$ is variable.

So, $A_{CD}$ *should* be a function of, $\left(0.4 - 0.2 * \frac{x}{L}\right)\left(0.4 - 0.2 * \frac{x}{L}\right)$

$$\int_{1.1}^{2.1} \gamma * \frac{A_{CD}}{9.8}\left(\left(\frac{x}{1}\right)^2\right)^2 dx = 22 * \frac{1}{9.8} * \int_{1.1}^{2.1} \left(0.4 - 0.2 * \frac{x}{L}\right)^2 x^4 \; dx$$

$$= 2.245 * \frac{2.1 - 1.1}{2} * ((0.4 - 0.2 * 1.8887)^2 * 1.8887^4 + (0.4 - 0.2 * 1.3113)^2 * 1.3113^4)$$

$$= 0.07004 \text{ kN}\frac{s^2}{m}$$

$$m^* = \int_0^{0.3} \mu_{AB}\left(\psi(x)\right)^2 dx + \int_{0.3}^{1.1} \mu_{BC}\left(\psi(x)\right)^2 dx + \int_{1.1}^{2.1} \mu_{CD}\left(\psi(x)\right)^2 dx$$

$$= 0.335\text{kN}\frac{s^2}{m} + 1.12\text{kN}\frac{s^2}{m} + 0.07004\text{kN}\frac{s^2}{m} = 1.52504 \text{ kN}\frac{s^2}{m}$$

So, $Natural\ Frequency,\ \omega = \sqrt{\frac{k^*}{m^*}}$

Here,

$$k^* = 8.799 * 10^8 \frac{\text{kN}}{\text{m}}$$

$$m^* = 1.52504\text{kN}\frac{s^2}{m}$$

$$\omega = \sqrt{\frac{8.799 * 10^8 \frac{\text{kN}}{\text{m}}}{1.52504\text{kN}\frac{s^2}{m}}} = \mathbf{24020.17\ rad}/\text{s}$$

# 3. Horizontal Stiffness and Natural frequency using ETABS

ETABS is an engineering software product that caters to multi-story building analysis and design. Modeling tools and templates, code-based load prescriptions, analysis methods and solution techniques, all coordinate with the grid-like geometry unique to this class of structure.[1]

## 3.1. Limitations

- ETABS only uses 1D elements (beam, column) and 2D elements (shell, slab, wall). It does not have 3D elements (solid).
- ETABS is not good for masonry structures (do not have control over element type); good for RCC structures.
- Single members with variable cross-sections cannot be defined using ETABS's built-in options.
- Triangular divide is not possible for Shear wall element.

## 3.2. Considerations

- Considering 2d elements for analysis (Shear wall).
- Modulus of elasticity is changed as given problem.
- For variable cross-section smaller segments are taken for relatively accurate results. This results in nodes that are not perfectly matched for different segments or meshes.
- Manual meshing is provided only as Q4 elements. Meshing was not properly aligned (no such option was found by us while doing the analysis). This could cause significant error in results.
- 1kN of horizontal point load is provided to measure displacement from which horizontal stiffness is measured.

## 3.3. Results from ETABS

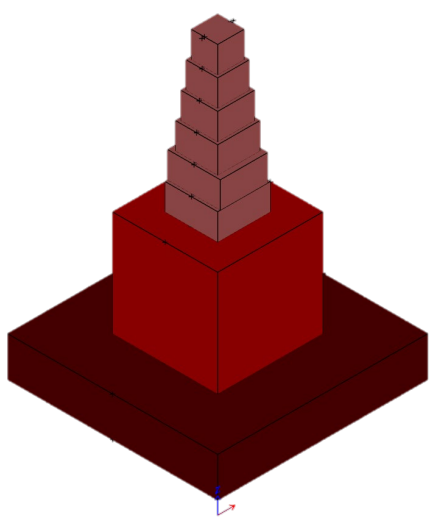


*Figure 3.1: 3D model in ETABS*

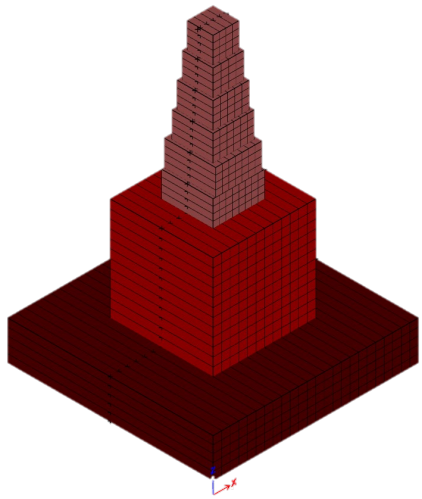


*Figure 3.2: 3D model after providing Q4 meshing*

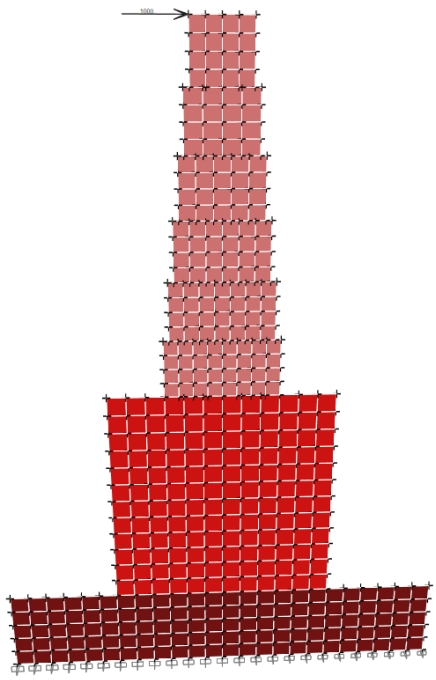


*Figure 3.3:Application of 1000N horizontal point load*

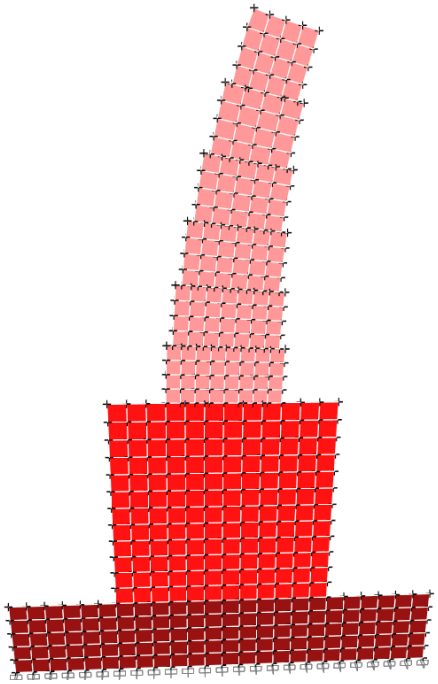


*Figure 3.4: Displacement due to horizontal loading*

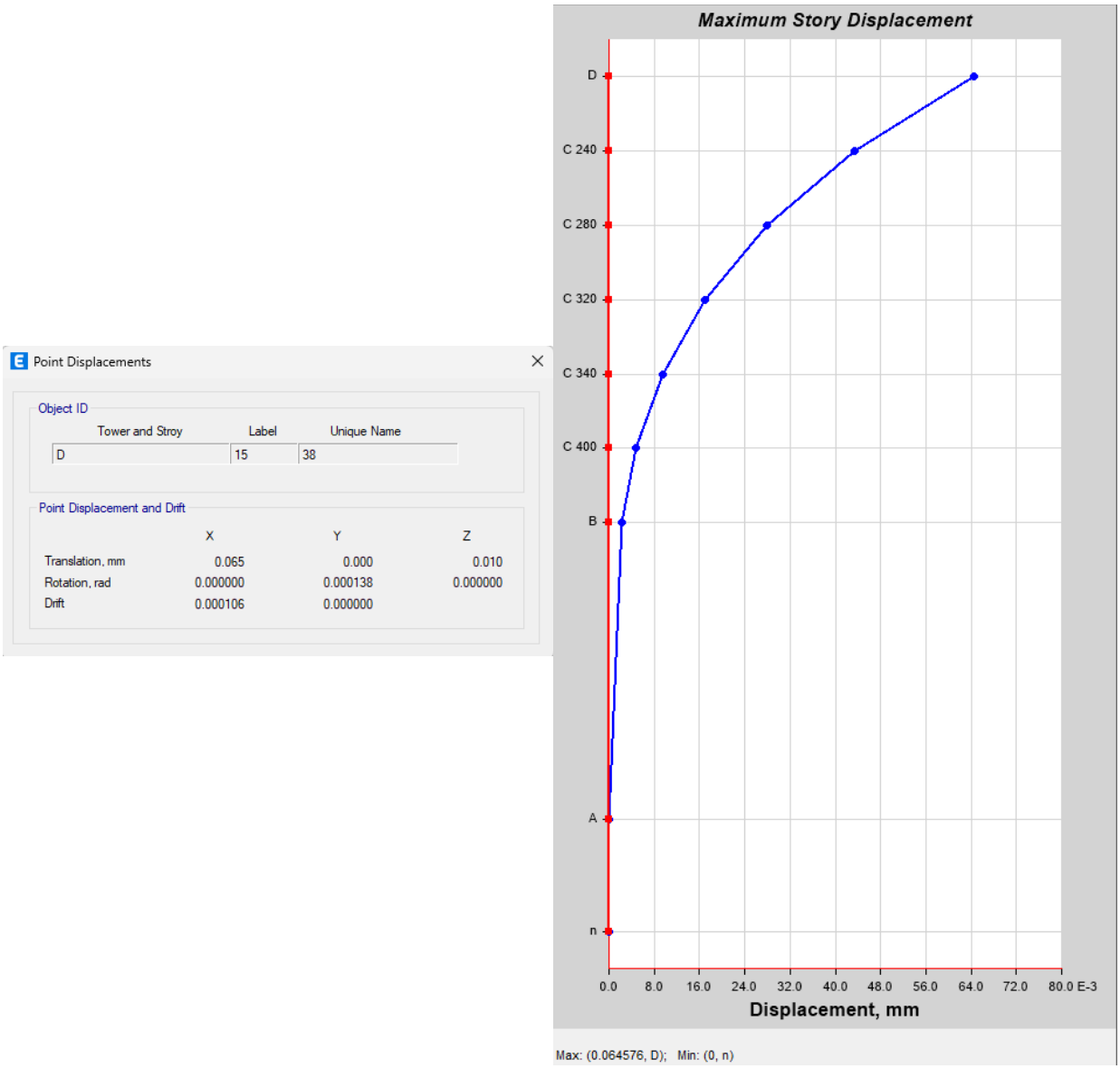



*Figure 3.5: Horizontal Displacement (ETABS)*

So, Horizontal Stiffness, $k^* = \frac{f}{u} = \frac{1kN}{0.065*10^{-3}m} = \mathbf{15.38 * 10^3 \frac{kN}{m}}$

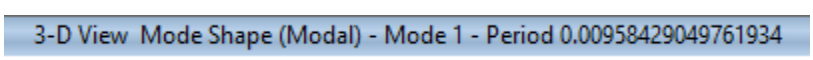



*Figure 3.6:1st Mode Time period (ETABS)*

Here, $T = 0.0095s$

So, Natural Frequency, $\omega = \frac{2\pi}{T} = \frac{2*\pi}{0.0095} = \mathbf{661.387 \frac{rad}{s}}$

# 4. Horizontal Stiffness and Natural frequency using SAP2000

SAP2000 is general-purpose civil-engineering software ideal for the analysis and design of any type of structural system. Basic and advanced systems, ranging from 2D to 3D, of simple geometry to complex, may be modeled, analyzed, designed, and optimized using a practical and intuitive object-based modeling environment that simplifies and streamlines the engineering process. The SAPFire® Analysis Engine integral to SAP2000 drives a sophisticated finite-element analysis procedure. An additional suite of advanced analysis features are available to users engaging state-of-the-art practice with nonlinear and dynamic consideration. Created by engineers for effective engineering, SAP2000 is the ideal software tool for users of any experience level, designing any structural system.[2]

## 4.1. Limitations

- SAP2000 is not good for masonry structures (do not have control over element type); good for RCC structures.

## 4.2. Considerations

- Modulus of elasticity is changed as given problem.
- Manual meshing is provided only as Q4 elements. Meshing was not properly aligned (no such option was found by us while doing the analysis). This could cause significant error in results.
- Triangular meshing is not explored.
- 1kN of horizontal point load is provided to measure displacement from which horizontal stiffness is measured.

## 4.3.Results from SAP2000

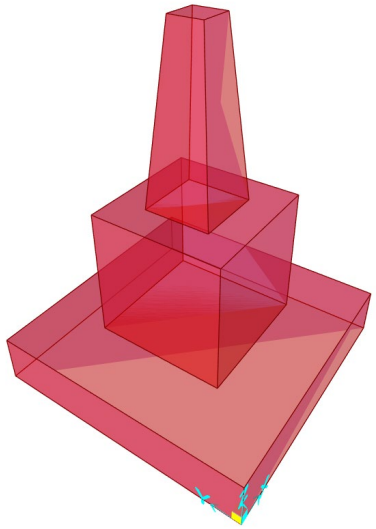


*Figure 4.1: 3D model in SAP2000*

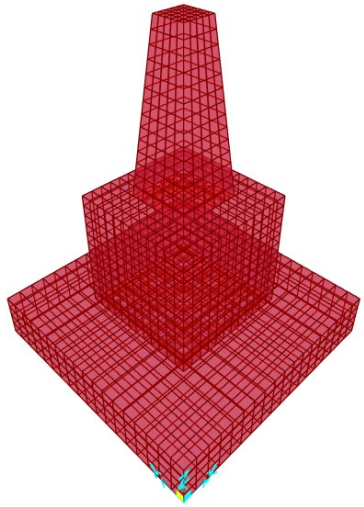


*Figure 4.2:3D model after providing Q4 meshing*

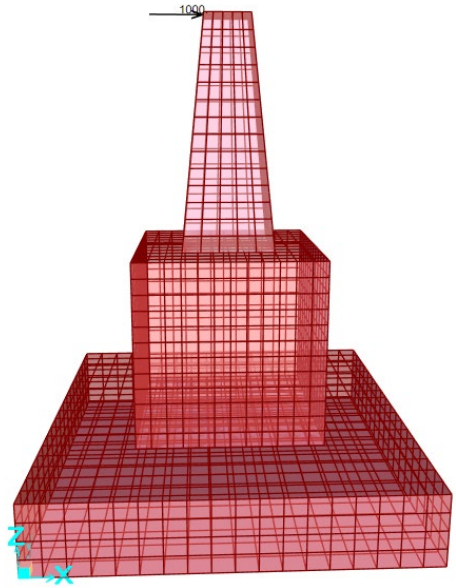



*Figure 4.3:Application of 1000N horizontal point load*

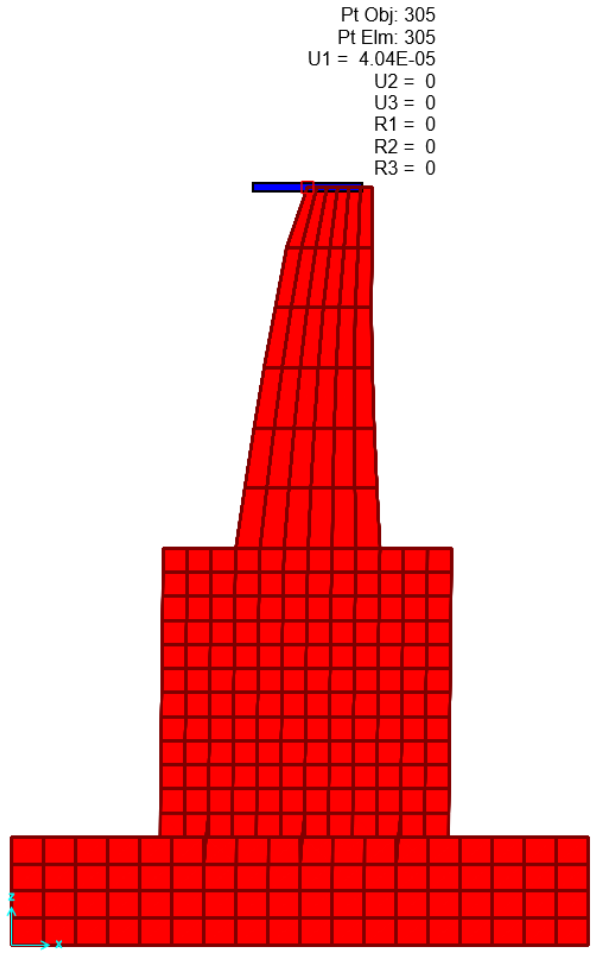
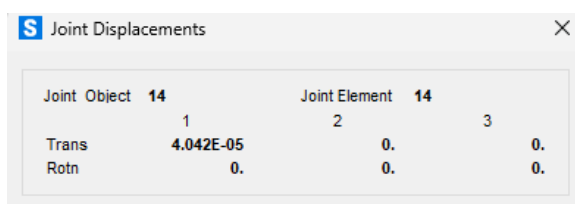



*Figure 4.4:Displacement due to horizontal loading*



Joint Displacements

| Joint Object | 14 | Joint Element | 14 |
|---|---|---|---|
| | 1 | 2 | 3 |
| Trans | 4.042E-05 | 0. | 0. |
| Rotn | 0. | 0. | 0. |

*Figure 4.5: Maximum displacement due to horizontal loading*

So, Horizontal Stiffness, $k^* = \frac{f}{u} = \frac{1kN}{4.042*10^{-5}m} = \mathbf{24.74 * 10^3 \frac{kN}{m}}$

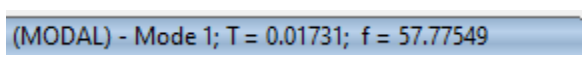



*Figure 4.6: 1st Time period*

Here, $T = 0.01731s$

So, Natural Frequency, $\omega = \frac{2\pi}{T} = \frac{2*\pi}{0.01731} = \mathbf{362.98 \frac{rad}{s}}$

# 5. Horizontal Stiffness and Natural frequency using Ansys Mechanical

Ansys Mechanical enables solving complex structural engineering problems and making better, faster design decisions. With the finite element analysis (FEA) solvers available in the suite, the software allows customization and automation of solutions for structural mechanics problems. It also facilitates parameterization to analyze multiple design scenarios. Ansys Mechanical is a dynamic tool with a complete range of analysis capabilities.[3]

## 5.1.Considerations

- Limited knowledge is used for using this software.
- Better triangular meshing was possible but was not done while doing the project.
- For Q4 elements some triangular elements also added automatically. This could be prevented but requires further knowledge on this software.

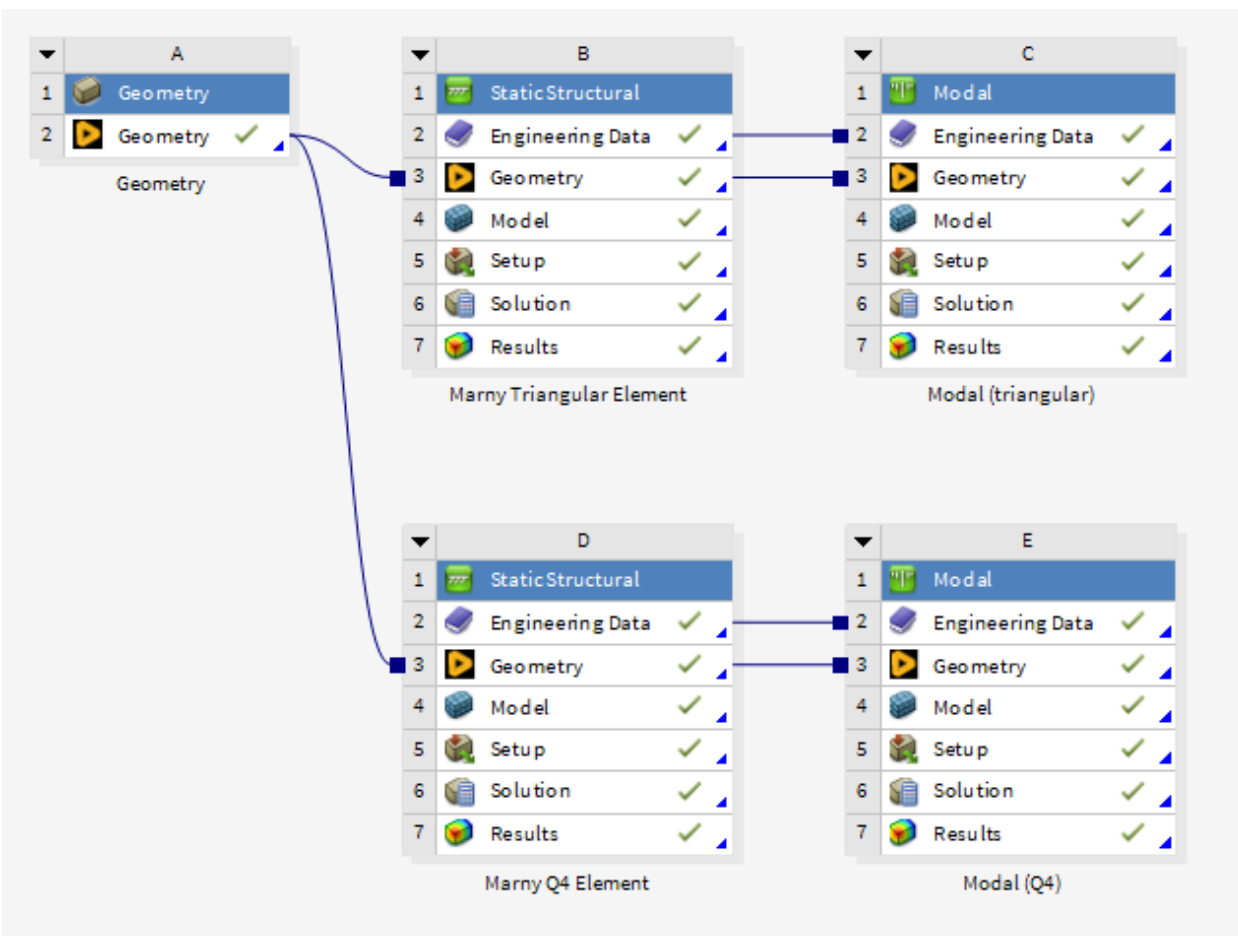



*Figure 5.1 Project Workbench in Ansys Mechanical*

| Property | Value | |
|---|---|---|
| Material Field Variables | Table | |
| Density | 2200 | kg m^-3 |
| Isotropic Elasticity | | |
| Derive from | Young's Mod... | |
| Young's Modulus | 10.85 | GPa |
| Poisson's Ratio | 0.22 | |
| Bulk Modulus | 6.4583E+09 | Pa |
| Shear Modulus | 4.4467E+09 | Pa |

*Figure 5.2: Properties used for masonry structure*

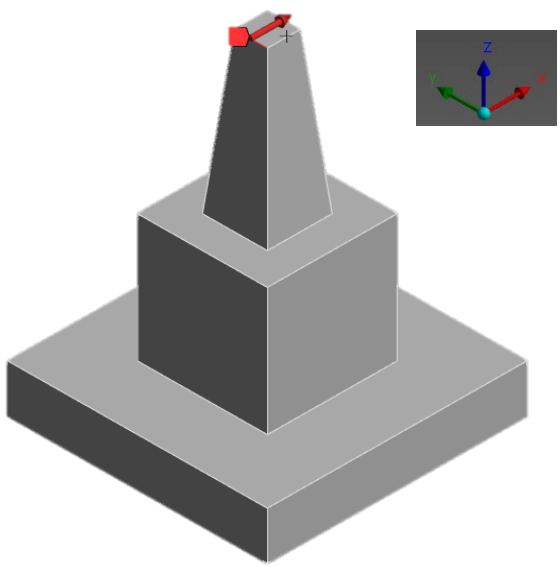



*Figure 5.3: 1kN load applied on the model*

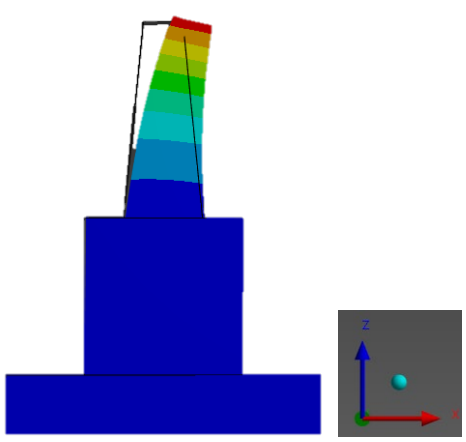



*Figure 5.4: Deformed shape after load application*

## 5.2. Many Triangular Elements

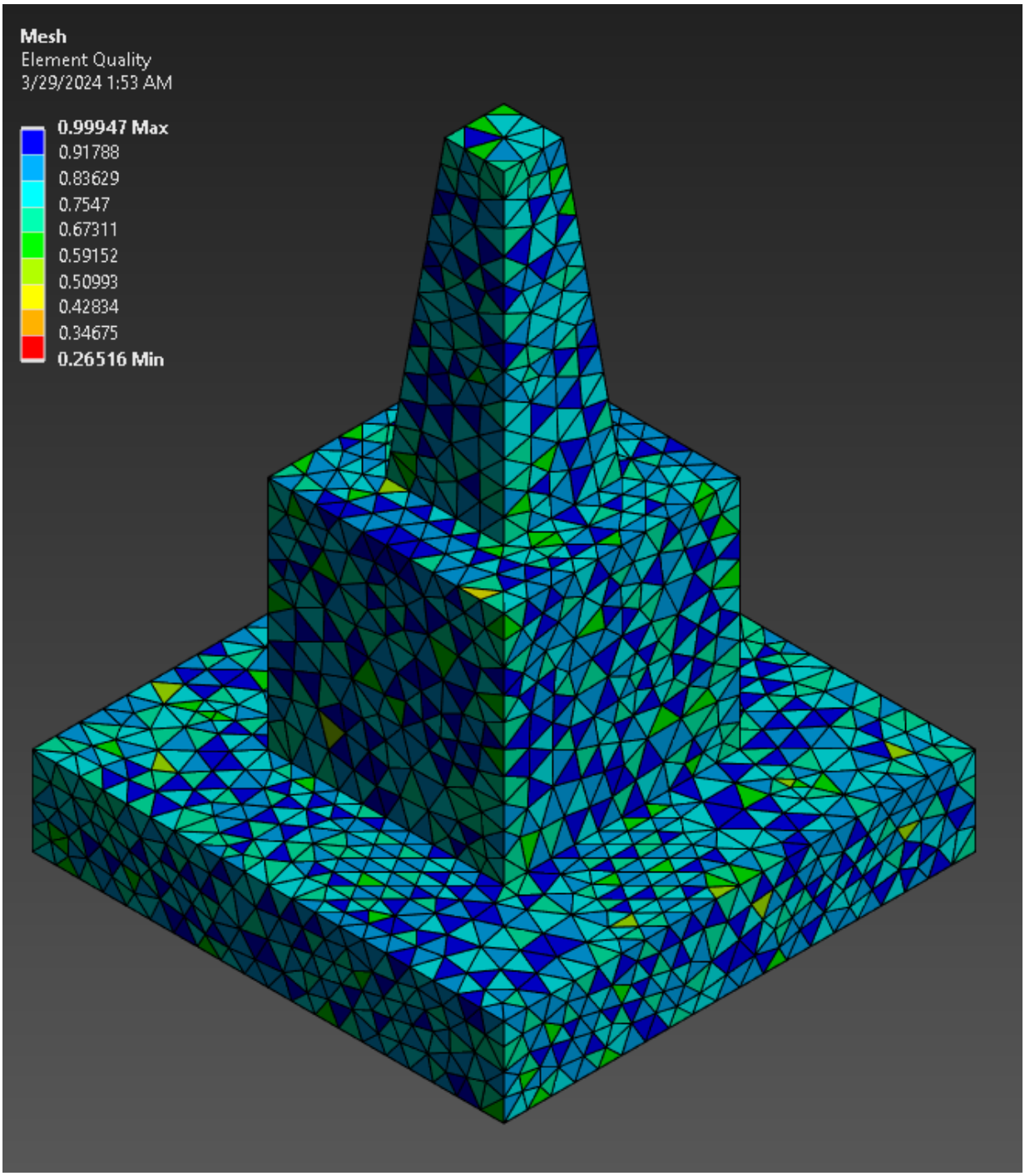



*Figure 5.5: Meshing with many triangular elements (a)*

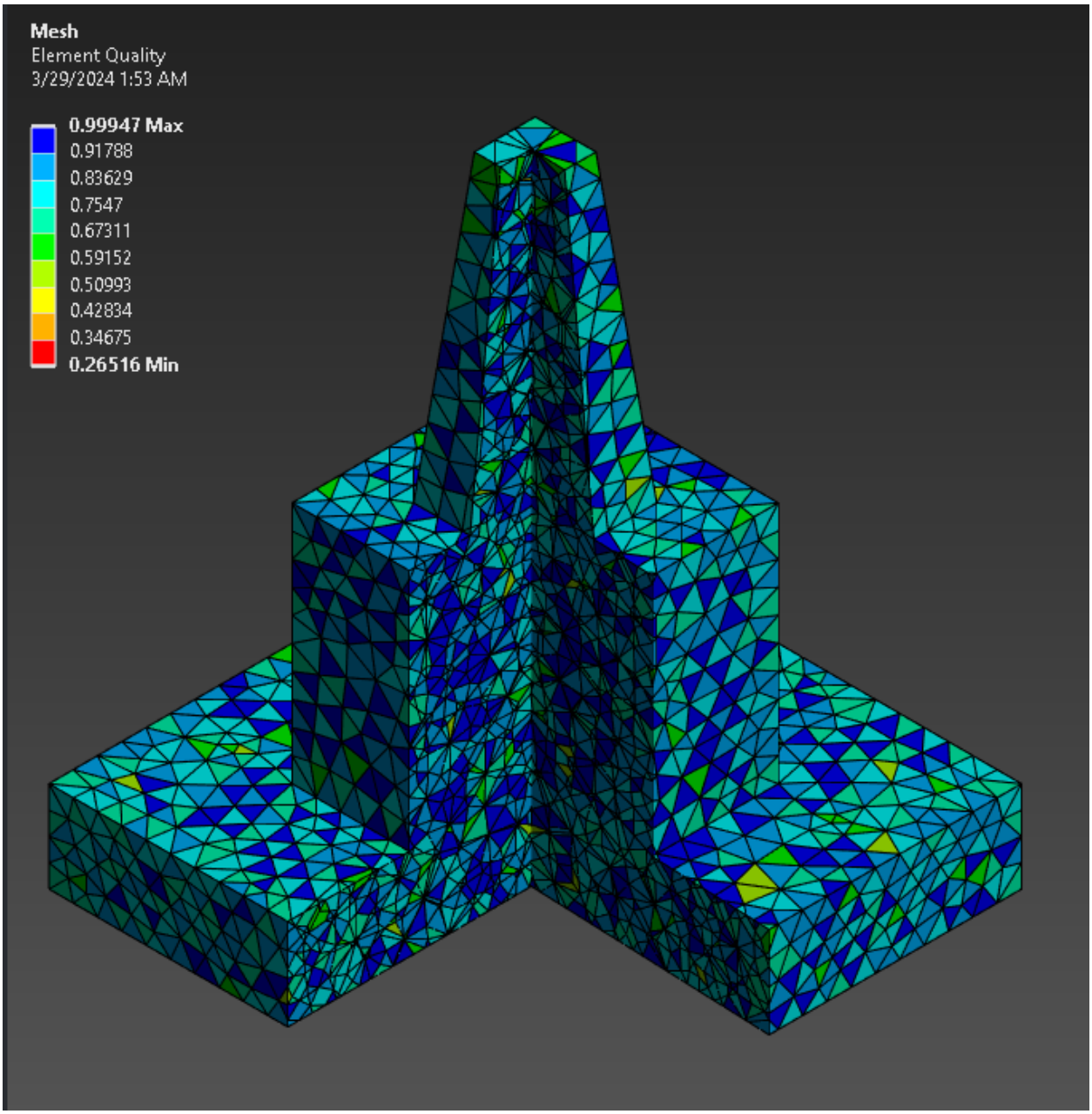



*Figure 5.6: Meshing with many triangular elements (b)*

| Statistics | |
|---|---|
| Nodes | 36242 |
| Elements | 24142 |

*Figure 5.7: Number of nodes and elements in many triangular elements meshing*

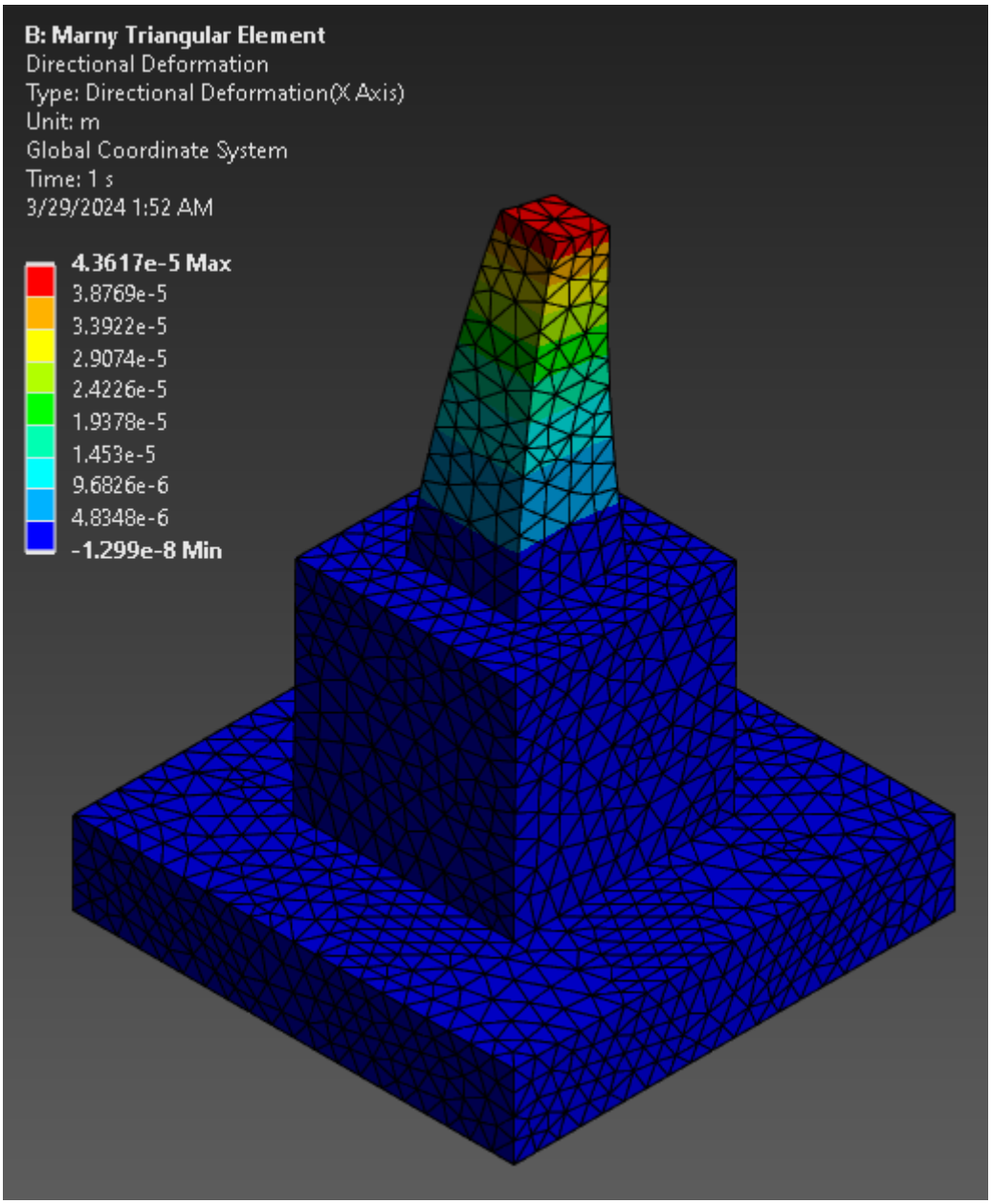



*Figure 5.8: Directional Deformation in many triangular elements*

So, Horizontal Stiffness, $k^* = \frac{f}{u} = \frac{1kN}{4.3617*10^{-5}m} = \mathbf{22.93 * 10^3 \frac{kN}{m}}$

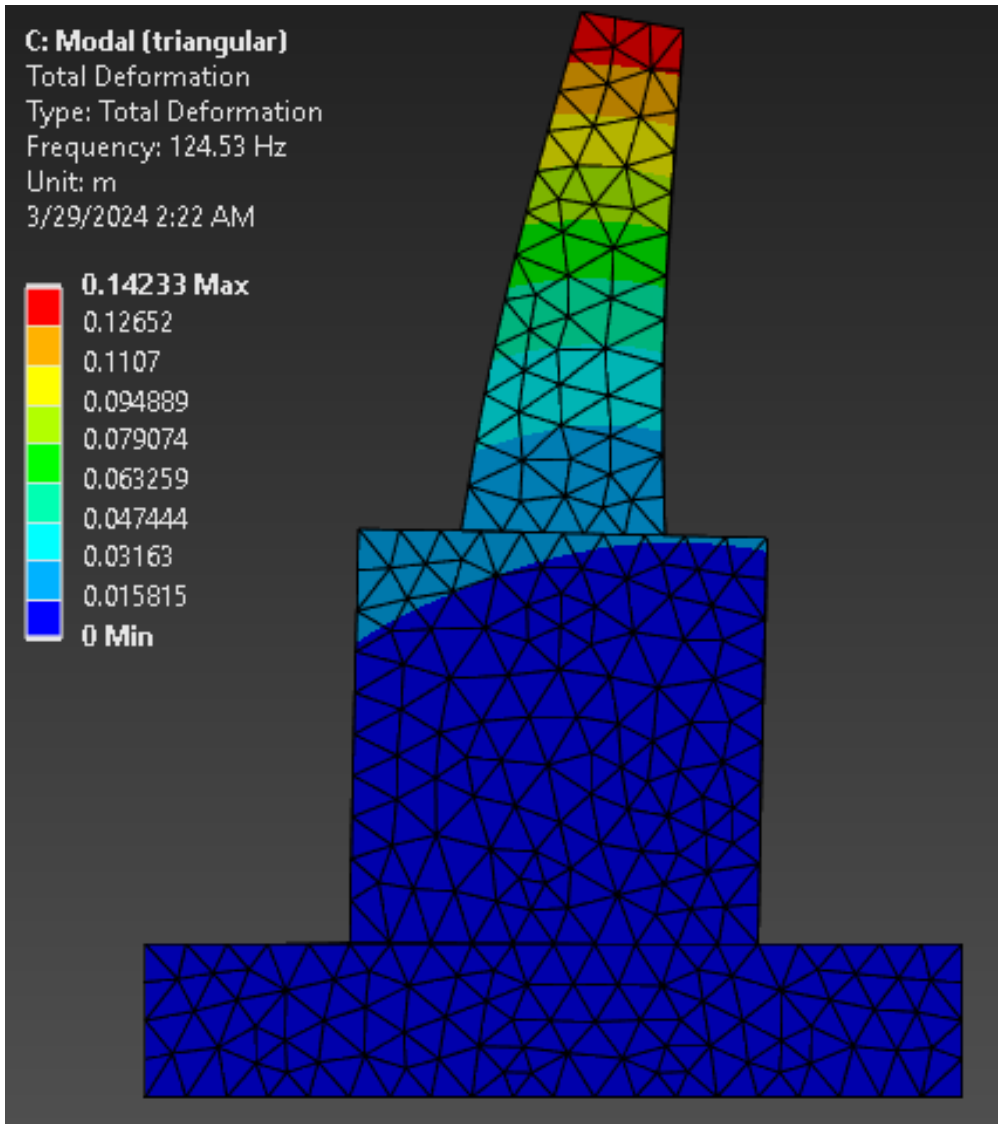



*Figure 5.9: 1st Modal shape and Natural frequency (triangular)*

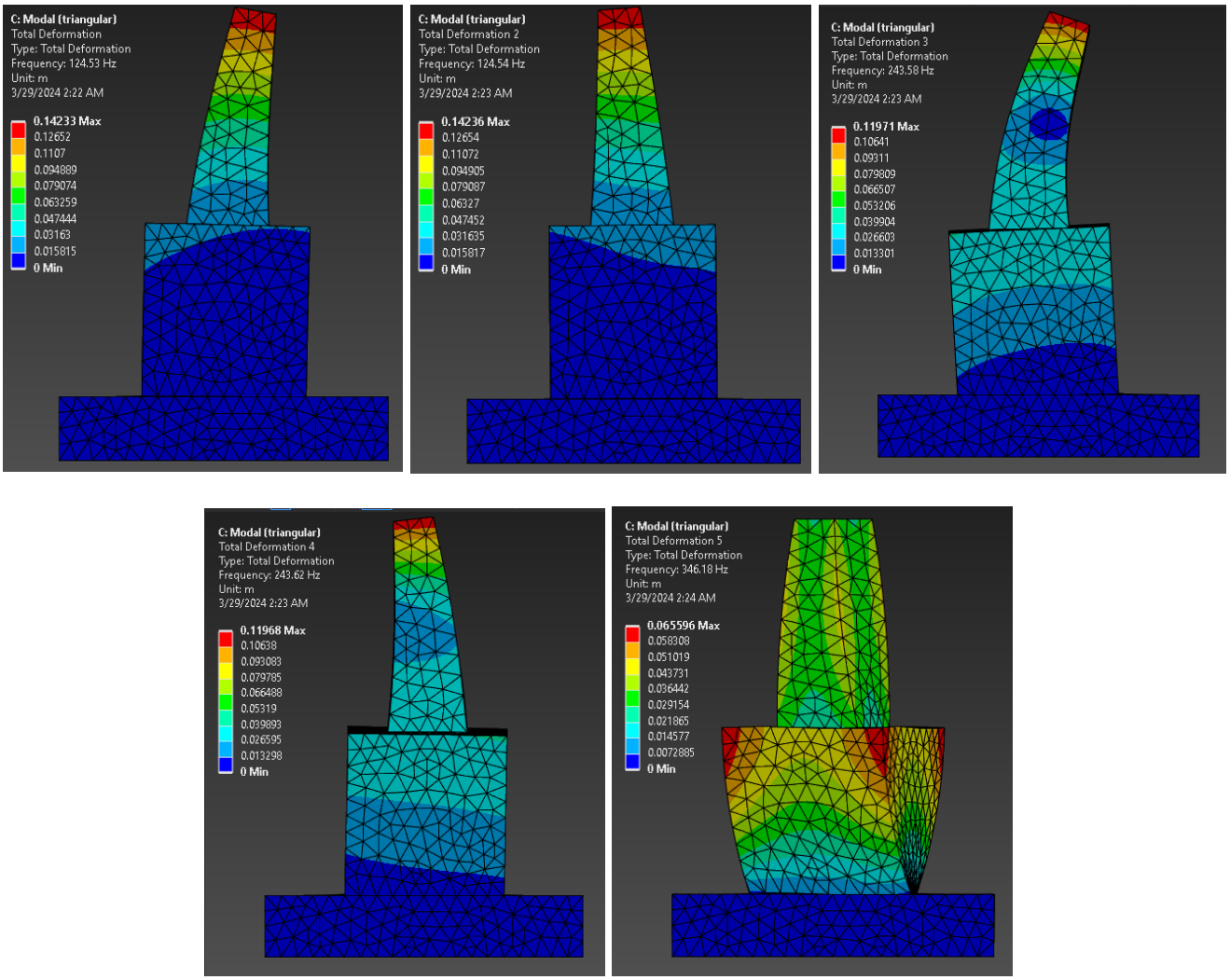



*Figure 5.10: Some other modes for many triangular elements*

So, 1st Natural Frequency $= 124.53Hz = 2\pi * 124.53 = \mathbf{782.44}\,\frac{\boldsymbol{rad}}{\boldsymbol{s}}$

## 5.3.Many Q4 Elements

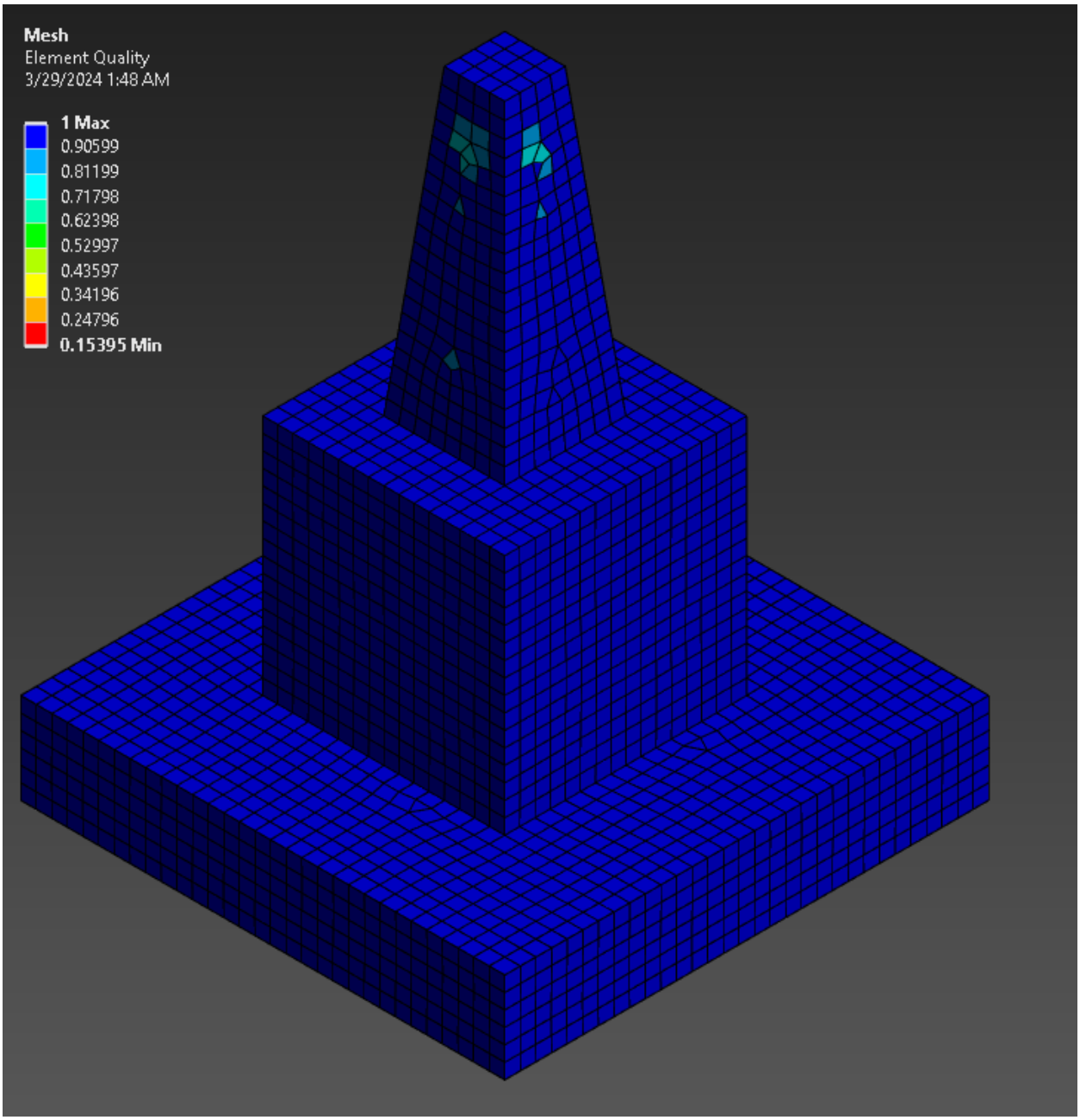



*Figure 5.11: Meshing with many Q4 elements (a)*

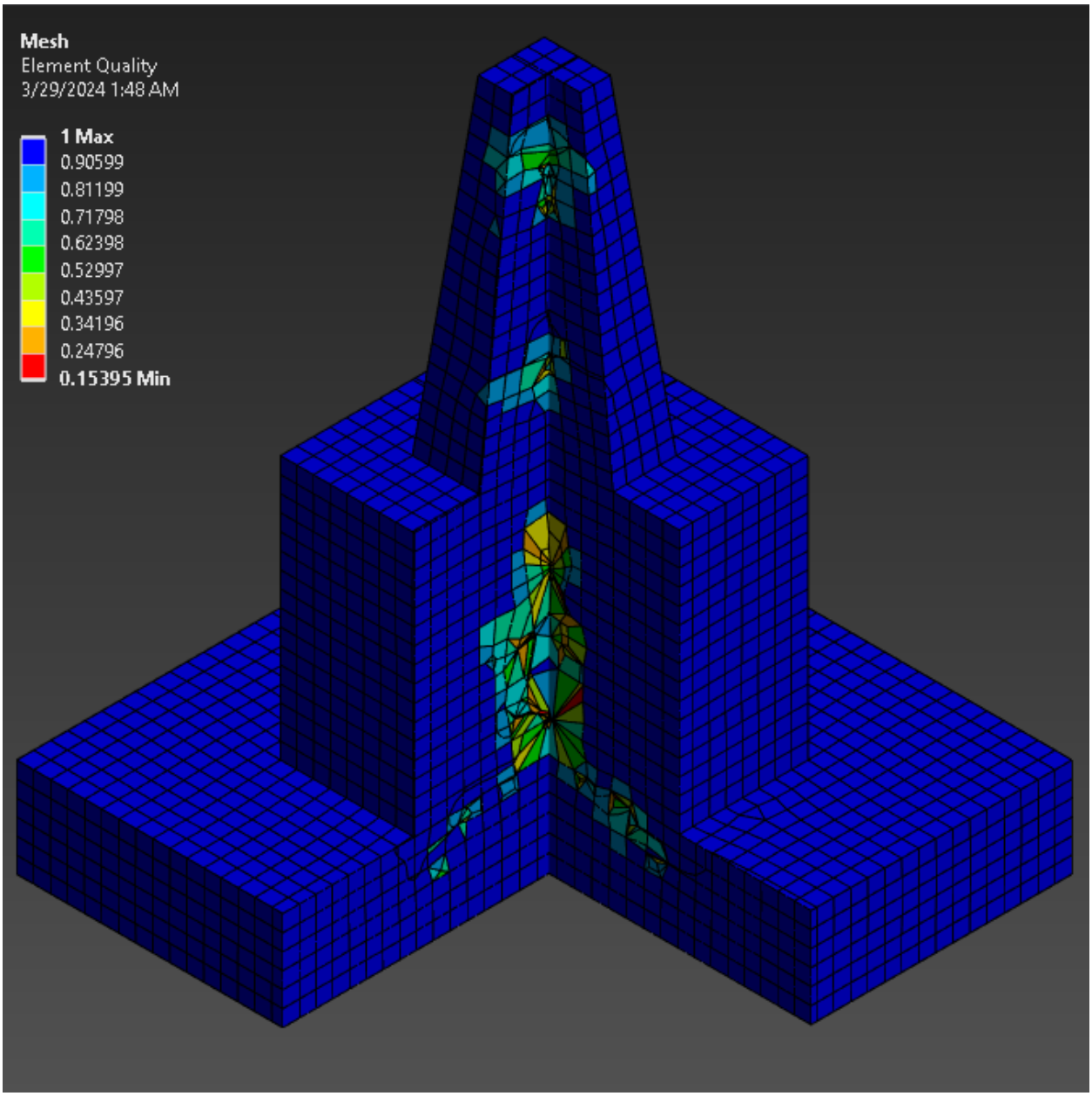



*Figure 5.12: Meshing with many Q4 elements (b)*

| Statistics | |
|---|---|
| Nodes | 49914 |
| Elements | 11282 |

*Figure 5.13: Number of nodes and elements in many Q4 elements meshing*

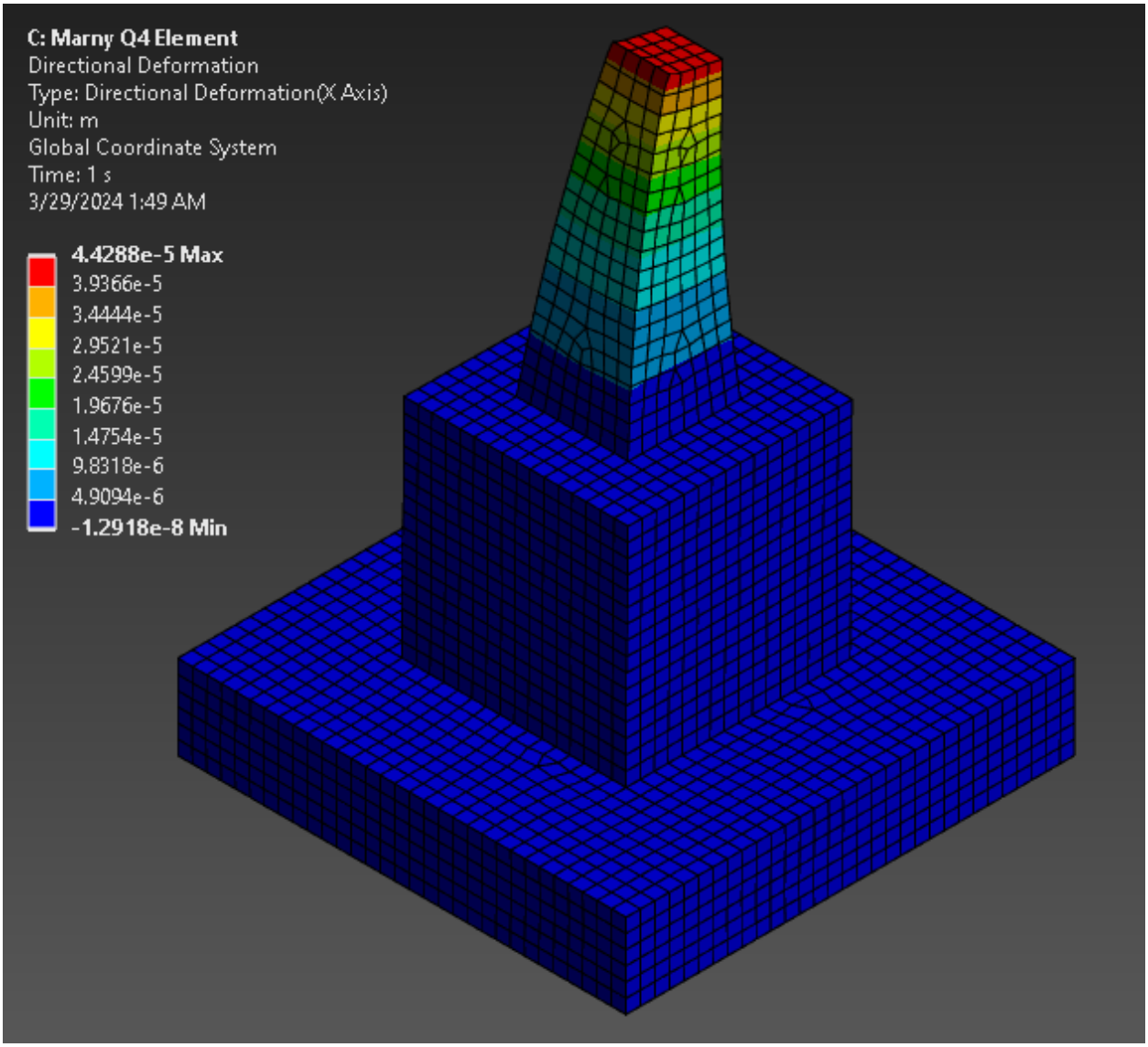



*Figure 5.14: Directional Deformation in many Q4 elements*

So, Horizontal Stiffness, $k^* = \frac{f}{u} = \frac{1kN}{4.428*10^{-5}m} = \mathbf{22.58 * 10^3 \frac{kN}{m}}$

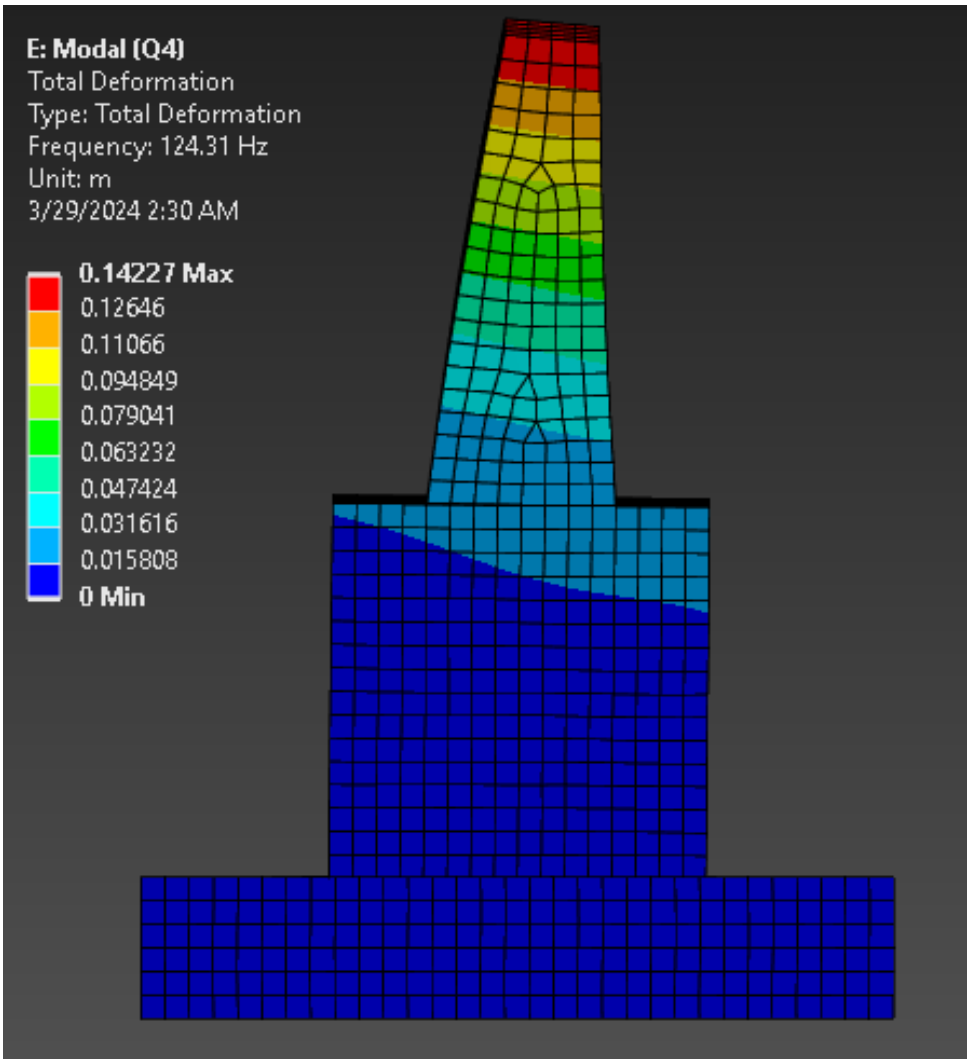



*Figure 5.15:1st Modal shape and Natural frequency (Q4)*

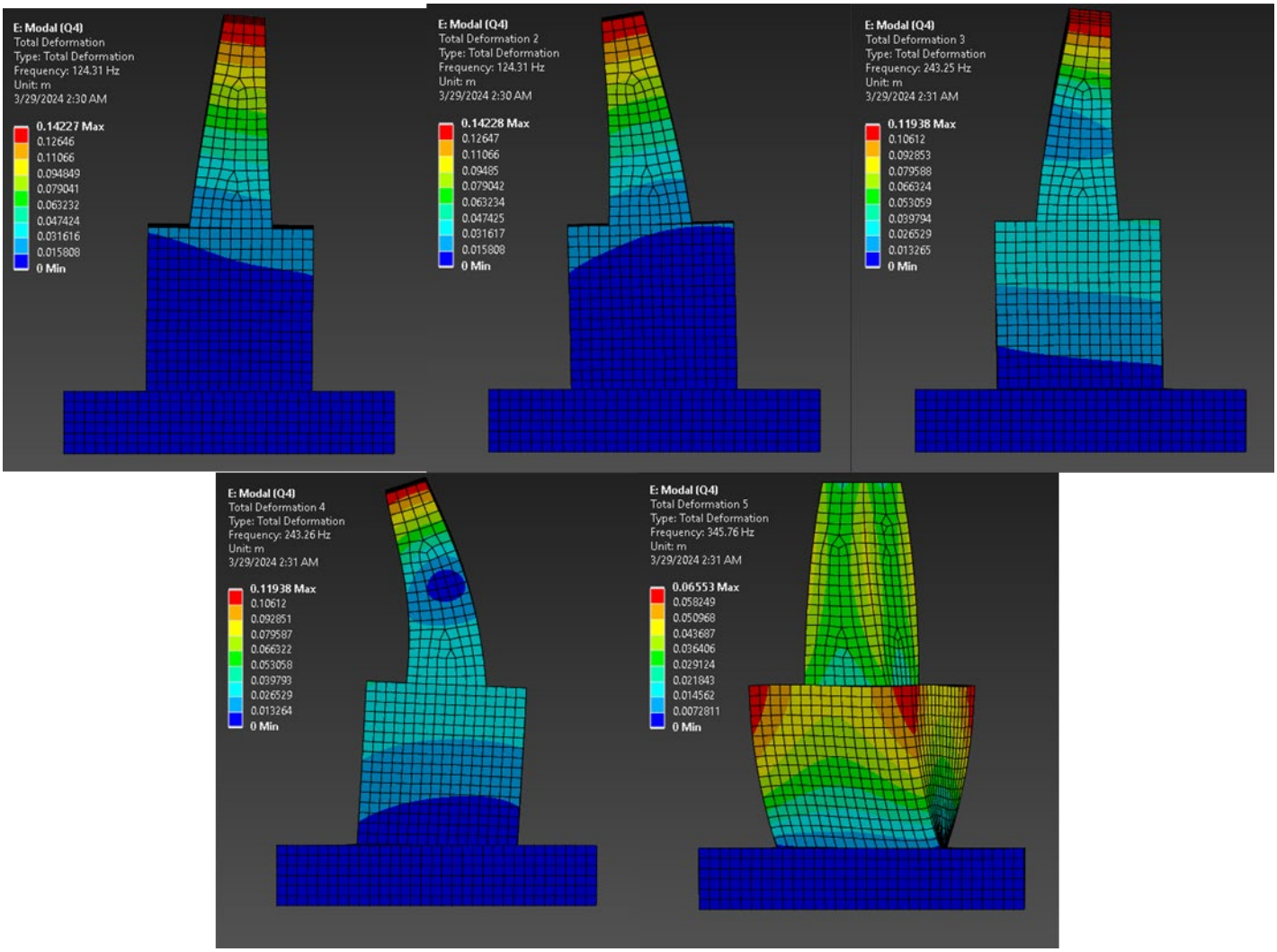



*Figure 5.16: Some other modes for many Q4 elements*

So, 1st Natural Frequency $= 124.31Hz = 2\pi * 124.31 = \mathbf{781.06}\,\frac{\boldsymbol{rad}}{\boldsymbol{s}}$

# 6. Comparison

*Table 6-1: Comparison Between Results*

| | Shape Function | ETABS | SAP2000 | Ansys (triangular) | Ansys (Q4) |
|---|---|---|---|---|---|
| **Horizontal Stiffness** | $8.799 * 10^9 \frac{\mathrm{kN}}{\mathrm{m}}$ | $15.38 * 10^3 \frac{kN}{m}$ | $24.74 * 10^3 \frac{kN}{m}$ | $22.93 * 10^3 \frac{kN}{m}$ | $22.58 * 10^3 \frac{kN}{m}$ |
| **Natural Frequency** | $24020.17 \frac{\mathrm{rad}}{\mathrm{s}}$ | $661.387 \frac{rad}{s}$ | $362.98 \frac{rad}{s}$ | $782.44 \frac{rad}{s}$ | $781.06 \frac{rad}{s}$ |

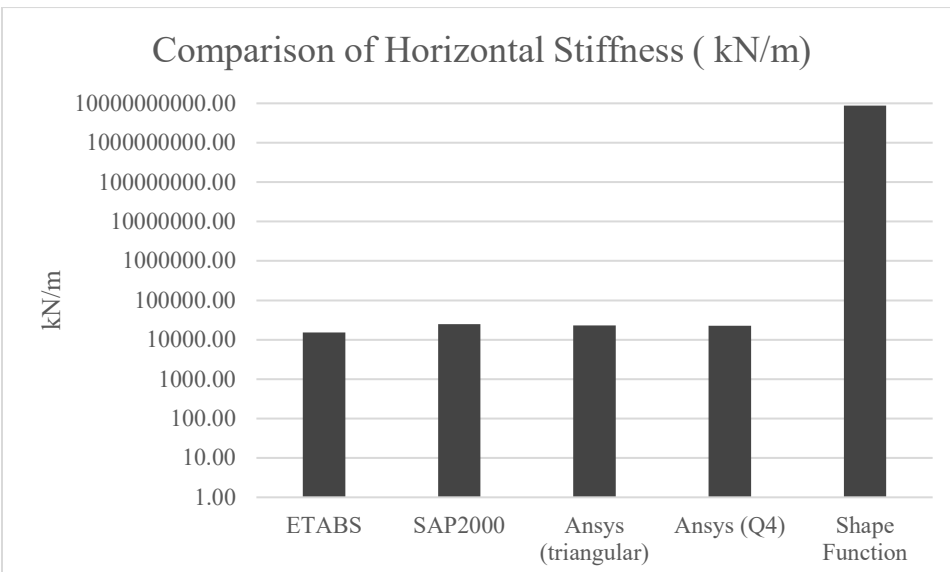



*Figure 6.1: Comparison of Horizontal Stiffness*

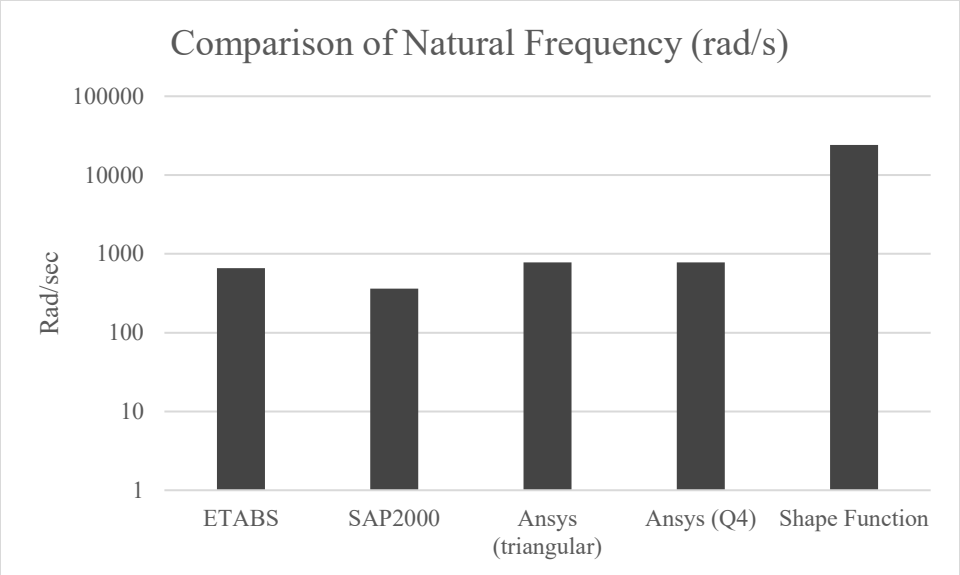



*Figure 6.2: Comparison of natural Frequency*

## 6.1. Possible reasons for such variations in results

- Assumed Shape function is not fully appropriate for the provided structure.
- There could be error in units and unit conversions while solving using shape function as well as inserting into to software.
- Meshing of the model is different for every software.
- New orientation with all the software.

# 7. References

[1] "Home - ETABS - Computers and Structures, Inc. - Technical Knowledge Base." Accessed: Mar. 27, 2024. [Online]. Available: https://wiki.csiamerica.com/display/etabs/Home

[2] "Home - SAP2000 - Computers and Structures, Inc. - Technical Knowledge Base." Accessed: Mar. 27, 2024. [Online]. Available: https://wiki.csiamerica.com/display/sap2000/Home

[3] "Ansys Mechanical | Structural FEA Analysis Software." Accessed: Mar. 29, 2024. [Online]. Available: https://www.ansys.com/products/structures/ansys-mechanical